

C-SQD Platform Specification

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1 Introduction

The current academic publishing system undermines scientific progress. Science advances through conjecture and rigorous criticism, yet traditional publishing models fail to reward the latter. This misalignment creates systemic problems at multiple levels of scientific inquiry.

1.1 The Problem

Academic journals profit from publishing exciting but often unreliable results, while scientists face pressure to produce positive findings or risk their careers. The critical work of peer review remains untransparent, uncompensated, and unrecognized in academic careers. This creates several fundamental problems:

1. **Misaligned Incentives:** Traditional journals profit from restricting access rather than improving reliability.
2. **Publication Bias:** The system incentivizes positive results over negative findings, regardless of scientific value.
3. **Devalued Criticism:** Peer review, essential to scientific progress, lacks recognition and compensation.
4. **Limited Transparency:** Review processes remain opaque, preventing community participation in critical evaluation.

Academic publishing represents the most lucrative segment of the global publishing industry. The top 5 conglomerates control 50% of market share, generating approximately \$2 billion in annual profit at 20-40% margins. The issue is not the magnitude of these profits but that they derive from monetizing academics' free labor while restricting knowledge access rather than enhancing reliability.

1.2 The C-SQD Solution

C-SQD is an integrated publishing platform built on the premise that market forces can be aligned with scientific progress. The platform combines:

- A publisher with transparent review processes
- A paid peer review system
- A scientific social network
- An AI assistant for scientific research

This integration creates a system where identifying errors and strengthening research quality becomes as valuable as publishing novel claims. Rather than paying to publish behind paywalls, manuscript fees directly fund large-scale public review processes where every identified flaw becomes a valuable contribution to the field.

1.3 Intellectual Foundations

Our goal is to elevate the status of peer review and to improve the reliability of institutional science. Doing so requires reconceiving the incentive structures at the heart of academic publishing.

Mediating Assessments Protocol

The platform’s design draws inspiration from Kahneman, Sibony and Sunstein’s Mediating Assessments Protocol (MAP), a framework introduced to reduce noise in judgment. Although bias in scientific evaluation receives considerable attention, noise, unwanted variability in assessments, remains underaddressed. MAP principles inform several C-SQD features:

- Replacing the initial accept/reject dichotomy with ongoing post-publication peer review
- Using multidimensional evaluation tuples (see section 3) rather than single aggregate metrics
- Employing independent assessments (*ElementReviews*) along different dimensions of research quality

Structured Review Framework

C-SQD implements two core review mechanisms:

ElementReviews (ERs) These focused evaluations assess manuscripts against specific criteria from the Common Research Weakness Enumeration (CRWE)—a comprehensive taxonomy of research vulnerabilities. The platform algorithmically generates *ElementReview* solicitations (ER-S) that match reviewers with manuscripts, specific CRWE criteria, and a payment scheme. Matching is based on the experience of the reviewer with respect to the CRWE criteria and the subject matter of the manuscript. Reviewers may also submit unsolicited *ElementReviews* (unpaid). This structure ensures a thorough multidimensional assessment while maintaining appropriate anonymity during review. While all *ElementReviews* are visible on the platform, they only become reidentified with some delay, e.g., 1 month after the initial public review phase ends.

SynthesisReviews *SynthesisReviews* are more analogous to traditional narrative reviews that evaluate the manuscript as a whole. After an initial public review period during which *ElementReviews* accumulate, the platform may solicit *SynthesisReviews* that integrate and contextualize the *ElementReviews*. This separation of broad initial review and expert synthesis combines crowd wisdom with structured expert evaluation.

1.4 Business Model Innovations

C-SQD’s innovation lies in creating incentive structures better aligned with science’s dual pillars, conjecture and criticism. Our model:

- Compensates reviewers for *ElementReviews* through manuscript submission fees
- Creates status incentives for expert *SynthesisReviews*
- Implements a review challenge system that incentivizes ongoing critical evaluation
- Establishes transparent and public reviewer records that recognize the true importance of scientific criticism
- Maintains open-access policy for manuscripts, however, reviews and other insights require subscription

Financial sustainability comes not from manuscript submission fees - which largely fund review - but from secondary revenue streams associated with the social network (e.g., review challenge fees, bug bounty fees, v-tags, nonstandard manuscript evaluations, etc.) and the AI assistant. This approach recognizes that while science requires funding—ultimately derived from taxpayers—the current allocation fails to optimize knowledge creation.

2 Review Process

The platform implements a three-phase review process that separates initial broad review from expert synthesis. This separation serves to combine the benefits of crowd wisdom with structured expert evaluation while maintaining appropriate anonymity at each stage.

Phase 1: Public Review

Double Blind Review (Author pays review fee)

The double-blind public review period is specific to each manuscript. Both authors and reviewers remain anonymous. This phase is open to all platform users, though only solicited reviews may receive compensation sourced from manuscript submission fees.

The C-SQD algorithm creates ElementReview solicitations (ER-S) in a stochastic manner. Each ER-S consists of a reviewer ID, manuscript, CRWE criterion, and **payment scheme**. The probability that a specific reviewer is matched with an ER-S depends on multiple factors including their demonstrated expertise in past reviews, publication record, and status as a platform subscriber. The payment scheme may be simple (e.g., guaranteed payment for accepting the ER-S and submitting the completed ElementReview) or more complicated and involve verification of some sort. Platform users who do not initially receive an ER-S can petition to be solicited for a particular manuscript.

When accepting an ER-S, reviewers commit to this ElementReview becoming part of their publicly accessible review record. In addition to their submission of ElementReviews, reviewers can submit petitions for other CRWE criteria to be applied to the manuscript. The algorithm uses these petitions when creating new ER-S assignments and constructing the manuscript evaluation tuple. This system balances two needs: biasing new ER-S creation towards potential problem areas while maintaining independence among reviewer judgments by not presupposing specific issues.

The public review period continues, generating new ER-S assignments until the allocated review funds are depleted. After closing, reviewers can still submit unpaid reviews and accrue status benefits from quality contributions. Additionally, reviewers or corporate entities can fund new review rounds or establish Bug Bounties.

Each reviewer can have at most $N_{pub-max}$ concurrent active ElementReviews. **Once all 3 review phases are completed, perhaps with some delay, ElementReviews (RE's) are reidentified and become part of each reviewer's public record.**

Non-anonymous Review (Author does not pay review fee)

Authors who do not pay for review may still publish their work on the platform. These manuscripts also enter a public review period with a maximum duration (e.g., 2 years). All review submissions are uncompensated. The platform does not advertise the manuscript, but it remains discoverable.

Phase 2: Revisions and author responses

During and after the public review period, the author can submit a response with revisions up to two times. They may accept or argue against aspects of the findings of the public review period. They may or may not include additional analyses, experiments, etc.; however, doing so may warrant a second public review period (and additional review fees). At the end of phase 2, the manuscript evaluation tuple will be available (see Section 3).

Phase 3: Expert Synthesis Review

Any manuscript subject to a solicited and / or unsolicited public review may reach a significance level that warrants expert SynthesisReview. This significance level is based in part on the manuscript's evaluation tuple at any time after the close of the public review period.

Following the public review period, algorithmically solicited expert reviewers craft SynthesisReviews that take into account the results of the public review period. During this phase, the authors of the manuscript and

the expert reviewers remain anonymous to each other if the public review phase was double-blind. Manuscript authors can specify preferences for reviewer communities from which the algorithm selects expert reviewers, although these preferences bias rather than constrain the selection process.

This phase is unpaid, but confers a high status. It is possible that the expert reviewer that accepts solicitation for a manuscript was also a reviewer in the public review phase.

Each expert reviewer can have at most $N_{expert-max}$ concurrent active `ElementReviews`. **Once all 3 review phases are completed, perhaps with some delay, expert `SynthesisReviews` are reidentified and become part of each reviewer’s public record.**

3 Manuscript Evaluation, E

The platform evaluates manuscripts through a tuple function:

$$E(P|R, T_{eval}) \rightarrow (N, M, P, L, C)$$

This function takes as input the manuscript P (including supplementary materials such as code and data), a specific reviewer community R , and T_{eval} , e.g., at the present time, or at the time phase 2 of the review process closed. It outputs a five-dimensional evaluation tuple where

- N measures reported non-ethical problems (higher values indicate more issues)
- M measures reported ethical problems (higher values indicate more issues)
- P quantifies the manuscript’s significance
- L represents the depth of review scrutiny
- C tracks citation impact

This multidimensional approach preserves important distinctions that would be lost in a single metric. For example, a manuscript might contain technical issues while still making significant contributions to its field. Authors can influence the depth metric L of the review directly by paying for a higher-tier review or indirectly by writing manuscripts that attract uncompensated reviews and high challenge rates.

Platform users can dynamically recompute these evaluations for different reviewer communities (possibly for additional fees) and set default reviewer criteria for their interface. This flexibility serves various needs; for example, hiring committees might focus on evaluations from specific subfields, while journalists could analyze how research reception varies across different communities over time.

4 Reviewer Network

4.1 Communities

The platform allows reviewers to form specialized groups based on research interests. For example, cancer researchers could create a community and supplement the CRWE with field-specific review criteria. All users can access public reviews, but the platform enables filtering evaluations by community membership.

Communities serve two purposes:

1. They enable specialized discussions within subfields while connecting to broader scientific conversation.
2. They support practical uses: hiring committees can assess candidates using reviews from bespoke combinations of relevant subfields, while journalists can track how different communities receive research.

4.2 How Communities Work: The Tag System

C-SQD uses tags to implement communities. Tags identify reviewer attributes such as field specialization, research methods, and community affiliations. This system offers two key advantages:

1. Users can filter reviewers by combining any tags (e.g., "neuroscience" + "electrophysiology").
2. The platform recalculates manuscript evaluations using only the selected reviewer group.

To understand how communities form: when multiple reviewers adopt the same tag (e.g., "computational oncology group"), they create a community without requiring special platform features. Examples of tag combinations:

- "computational oncology" + "machine learning" finds reviewers working at this intersection.
- "theoretical physics" + "experienced reviewer" locates senior physicists with strong review records.

How Tags Work in Practice

Each reviewer's profile contains tags showing their expertise areas, methods, and community affiliations. Reviewers add their own tags without restrictions.

When viewing a manuscript, users can select tags to filter reviews through a specific "community lens." The platform then:

1. Finds all reviewers matching these tags.
2. Recalculates the manuscript's evaluation using only these reviewers.

Communities emerge naturally as tags gain adoption rather than through formal creation processes. Popular tags become de facto communities, while unused tags fade away over time.

Verified Tags

Platform users may establish *verified tags* (v-tags). Communities associated with a v-tag may decide on membership criteria, and there is an associated membership fee payable to the platform.

4.3 Professional Relationships

The platform tracks two types of professional connections:

1. **Collaboration networks:** Form automatically when reviewers publish joint reviews. Each reviewer joins the others' networks.
2. **Academic lineage:** Form when researchers establish mentor-mentee relationships. For example, a postdoc can request their advisor to confirm their academic relationship.

These relationship structures help users understand reviewer perspectives and navigate the scientific community more effectively.

5 Reviewer Status

During the platform's initial grace period, requirements for attaining reviewer status are relaxed to facilitate building the user base. After this period, reviewers must meet one of two criteria: receive an endorsement from a reviewer who has completed at least one expert SynthesisReview, or receive x endorsements from reviewers who have each completed at least y ElementReview solicitations.

All endorsements must specify the fields to which they apply, and they become public information displayed on both the provider's and receiver's profiles. This transparency helps maintain the integrity of the endorsement system while building visible professional networks.

5.1 Reviewer Evaluation

Reviewers that accept ER-S's by the platform algorithm, or solicitations for double blind expert synthesis are encouraged to complete their reviews (or notify the platform they are unable to complete the review) within a timely period - i.e., within 3 weeks. Reviewers who submit between 3-5 weeks are moderately penalized by the algorithm for future solicitation.

Reviewers who do not submit within 6 weeks are more severely penalized, as are reviewers that violate the ethical code of conduct (e.g., misrepresenting the manuscript in a malicious manner, engaging in ad hominem attacks against the author, etc.).

6 The Challenge System

The challenge system addresses at least two problems related to the production of high-quality reviews.

- There may be very many ElementReviews and/or expert SynthesisReviews submitted for a particular paper. It is helpful to maintain a featured subset as the 'best' representatives for the purpose of quickly determining the most important kinds of critique a work has been subject to and how these change over time.
- Reviewers are already incentivized to contribute reviews via compensation. They are disincentivized from submitting obviously poor or misleading reviews due to the fact that all reviews become part of their public record. However, we also need incentives for the continued exploration of conflicts between rival critiques.

The platform allows reviewers to initiate either:

- Direct Challenges: Challenge an existing featured ElementReview or SynthesisReview. If the challenger's review wins, their ElementReview or SynthesisReview replaces the losing review as part of the featured subset.
- Petition Challenges: Publicly petition for another reviewer's ElementReview or SynthesisReview replacing an existing featured review of the same type.

Initiating a direct challenge incurs a modest fee. The purposes of this fee are to discourage excessive or frivolous challenges, and to sustain the financial health of the platform. Petition challenges incur no fees.

How are challenges evaluated?

It is important to note that there may be more than one challenger to a featured ElementReview or expert SynthesisReview. When a challenge is initiated, the challenger must select an election date. Successive elections are between two reviewers, the current incumbent and the challenger. Suppose a reviewer challenges an incumbent's featured review and sets the election date, D , which is 2 weeks into the future. If the incumbent loses another challenge before D , this election is canceled, and the challenger's fee becomes a credit that can be applied to a future challenge.

Votes are counted from petition challenges as well as from any platform users who happen to come across the in-progress challenge and cast a vote. The platform enables users to easily view a list of all the challenges associated with manuscripts or reviews with which they have previously participated.

A few rules:

- All challengers are identified.
- Challengers must set an election date at least 1 week into the future and on a date that does not overlap with existing challenges for the same featured review.
- Voters may change their votes as many times as they wish. Votes at the time of the election determine the winner.

- Voters can inspect all previous losing challenges submitted for the manuscript.
- There can be at most m challenges to a given ElementReview or SynthesisReview within the last 30 days.
- Each vote is weighted based on alignment between the vote caster’s review record and the featured review in terms of demonstrated expertise.
- Former incumbents cannot rechallenge within 30 days.
- If the incumbent previously won an election, any votes received during the previous election default to the incumbent in their next election.

7 How manuscript submission fees are allocated to pay for review

A viable approach that blends multiple mechanisms is to offer each manuscript both a guaranteed baseline of paid review coverage and the opportunity to attract additional paid ER-S’s if it garners sufficient community engagement. The following is an outline of this hybrid model.

1. **Guaranteed Minimum Allocation.** A fixed percentage of each manuscript submission fee (e.g. 30—50%) is reserved exclusively for paying reviewers who submit valid ElementReviews (*ER-S*) on that manuscript. This ensures that every manuscript benefits from at least a baseline number of funded reviews.
 - *Rationale:* Alleviates author concerns that their fees might not translate into meaningful review coverage for their own work.
 - *Implementation Detail:* The platform could define a baseline review target (e.g. 3 ElementReviews) and use the reserved funds to compensate reviewers who accept *ER-S* assignments until the baseline is met.
2. **Flexible Pool.** The remaining portion of the submission fee goes into a central pool. Any unused portion of the manuscript’s reserved funds (e.g., if fewer solicited reviewers than expected accept *ER-S* offers within say 2 years) can also flow back into this pool.
 - *Rationale:* Grants the platform the flexibility to distribute extra review resources where they are most needed, beyond each manuscript’s baseline coverage.
 - *Implementation Detail:* This pool is managed by an algorithm that weighs factors such as the total volume of submissions, the available group of reviewers and signals of community interest.
3. **Engagement-Triggered Allocations.** To the extent that a manuscript garners high community engagement—such as ElementReview solicitation (ER-S) petitions for existing or additional *CRWE* criteria, multiple ElementReview *challenges*, etc.—while it also receives disproportionately low unsolicited review, the platform algorithm allocates additional funds from the flexible pool to support deeper or more specialized reviews.
 - *Rationale:* Aligns with the goal of directing greater attention to manuscripts that the community deems especially important or complex.
 - *Implementation Detail:* The platform could use thresholds based on the number of *review petitions* or a ratio of engagements (*comments, upvotes, bug bounty proposals, etc.*) to trigger extra reviewer solicitations with higher payments.
4. **Per-Manuscript Cap.** To avoid a small number of popular or hyped submissions monopolizing the flexible pool, the platform enforces a rule such as:

Total funds allocated from the pool for any given manuscript $\leq k \times (\text{author’s submission fee})$.

- *Rationale:* Maintains fairness by preventing runaway consumption of reviewer funds by a select few manuscripts.
- *Implementation Detail:* The cap k might be initially set (e.g., $k = 2$ or 3) and then adjusted based on platform data and community feedback.

8 Incentivizing Authors to opt for Paid Review

Helps with Visibility The platform provides users with a customizable "headlines" feed that presents them with new (to them) and/or significant manuscripts. The algorithm for determining which manuscripts appear is user-customizable, but depends on the manuscript evaluation tuple. Authors therefore have an incentive to improve the review depth criterion for their manuscripts, and paying for review is an important means of doing so.

Review Customization. One powerful motivator for authors to pay the review fee is allowing them to petition for additional or specialized review criteria to be applied to their submission, such as rigorous reproducibility checks, methodological correctness evaluations, or even domain-specific validation for code and data.

Another motivator for authors to pay the review fee is the granting of **choice of the reviewer pool**: Authors who pay might exercise more control over the composition of reviewers the platform algorithmically solicits.

Long-Term Manuscript Support. Another incentive is to offer extended or ongoing review support for manuscripts whose authors pay for the formal review process. For example, the platform might provide them with partial fee waivers for challenges. This ongoing support can also include a mechanism for acknowledging and integrating major corrections or addenda, ensuring the continuous improvement of published research.

Social or Gamified Incentives. Finally, authors who pay for review may receive special social recognition within the platform ecosystem. For example, the system could display a distinctive badge indicating that a manuscript underwent a premium review path, or bestow on the author a higher-level status (e.g., "Gold Publisher") for consistently investing in rigorous evaluation. These visible markers build an environment in which paying for review confers social capital and professional prestige, attributes that are typically valued among researchers, hiring committees, and funders. The gamified element can go one step further: authors might unlock access to exclusive networking events, Q&A sessions, or cross-disciplinary forums that are reserved for those who have demonstrated a commitment to high-level peer review. By embedding these social signals into the platform, paying for review becomes more than a simple transaction; it becomes a recognized investment in quality science and an important differentiator of one's publication record. This is a potent incentive, as it leverages the human desire to build reputation and recognize the community, providing a clear structure that rewards authors who opt for the enhanced review track.

9 AI Utilization

Platform Tools

AI can help the platform understand which scientific problems its users are interested in.

This is relevant to:

- Search: Scientific problem-based manuscript search as a potentially more useful alternative to academic field or keyword based search
- Feed: Personalized user feed algorithms for manuscript and review discovery

AI can function as a creative assistant for academic research.

This would be a paid service made available to platform users. Broadly, the idea is to retrain or fine-tune an existing LLM with platform data including manuscripts, code, reviews, and supplementary research data. This would allow researchers to more easily understand what has been done before, to discover non-obvious connections between seemingly unrelated ideas, and more generally to solve a wide range of practical problems efficiently.

Example: A computational biologist is modeling immune selection. She is interested in testing the hypothesis that a person’s HLA profile and specifically its degree of homozygosity has a causal influence on the likelihood of tumor clone immune escape after treatment with anti-PD-L1 immune therapy. This is a complex hypothesis that can be addressed from many angles and with multiple data types. The rate of progress in this area of cancer research could be greatly improved if the AI assistant was aware of existing theoretical approaches that relate to the hypothesis, as well as published methods, data and code.

9.1 User Tools

AI tools are research products just like any other research product. They should also be subject to ongoing peer review.

Reviewers may utilize AI tools that have been sufficiently peer reviewed to supplement their reviews. The platform’s ethical code of conduct will require them to acknowledge when they do so. Reviewers who utilize AI must provide as part of their review a separate justification for the suitability of the specific AI tool to the manuscript.

Reviewers can develop AI suited to specific review tasks and make them available to others for a fee payable to the creator.

The platform can also run AI packages created by the community on large collections of manuscripts for a fee. Community-created AI packages must meet minimal standards including specifications of the kinds of manuscript (including manuscript data and code) the tool is appropriate for.

10 Platform Growth Strategy

C-SQD employs three strategic initiatives to accelerate adoption: the Referee Campaign, targeted Mini Grants, and the Conference Partnership Program.

10.1 The Referee Campaign

Referee serves as the entry point for manuscripts first presented elsewhere—on preprint servers or in traditional journals. These manuscripts can be uploaded, discovered, and reviewed on the C-SQD platform.

10.2 Strategic Network Engagement

Scientists collaborate within small, overlapping networks formed through shared academic experiences such as graduate school, postdoctoral fellowships, or academic lineage. C-SQD accelerates adoption by engaging key networks across multiple disciplines.

The platform partners with principal investigators (PIs), offering small discretionary grants to their labs in exchange for participation. Each PI commits to three actions:

- Publishing at least one modular manuscript (a paper focused on a small set of scientific problems or negative results) on C-SQD for each submission to a traditional journal
- Citing this modular manuscript in their submission to the traditional journal
- Providing a minimal level of peer review on the platform

This creates mutual benefits: PIs gain additional publications and lab funding, while C-SQD overcomes early-stage network effects and drives community adoption.

10.3 C-SQD Conference Partnership Program

The Conference Partnership Program streamlines the publication and review process for conference-associated manuscripts. Conference organizers pay a single fee to gain waived platform fees for all manuscripts published under their conference banner.

Upon program enrollment:

- Conference attendees add a specific conference tag to their reviewer profiles
- This tag remains exclusive to attendees, ensuring only eligible reviewers match with conference manuscripts
- The C-SQD platform manages the reviewer-manuscript matching process
- Reviewers receive ElementReview assignments based on their submitted manuscripts, community tags, and organizer-provided criteria

Conference organizers determine review anonymity levels before and during the conference. All reviews become public afterward, aligning with C-SQD’s standard policy. Organizers may require attendees to contribute a predetermined number of ElementReviews, promoting active participation in peer review and maintaining publication quality.

Only manuscripts meeting conference-defined criteria qualify for expert SynthesisReview. All submitted reviews, both ElementReviews and SynthesisReviews, enter reviewers’ public records, providing transparency, and acknowledging reviewer contributions. Conference manuscripts receive Digital Object Identifiers (DOIs) like any other C-SQD publication.